RAJALAKSHMI ENGINEERING COLLEGE

Santhosh S (dob 14.02.2008) 251801299-2025-AIDS

S2

Started on	Saturday, 22 August 2026, 1:40 PM
State	Finished
Completed on	Saturday, 22 August 2026, 1:48 PM
Time taken	8 mins 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    int i = 1;
    count++;

    int s = 1;
    count++;

    while (s <= n)
    {
        count++;
        i++;
        count++;
        s += i;
        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Santhosh S (dob 14.02.2008) 251801299-2025-AIDS

S2

Started on	Saturday, 22 August 2026, 1:50 PM
State	Finished
Completed on	Saturday, 22 August 2026, 1:56 PM
Time taken	6 mins 52 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    if(n == 1)
    {
        count++;
    }
    else
    {
        int i = 1;
        count++;

        for(; i <= n; i++)
        {
            count++;

            int j = 1;
            count++;

            for(; j <= n; j++)
            {
                count++;
```

```
29                  count++;
30                  break;
31              }
32
33              count++;
34          }
35
36          count++;
37      }
38
39      printf("%d", count);
40
41      return 0;
42  }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Santhosh S (dob 14.02.2008) 251801299-2025-AIDS

S2

Started on	Saturday, 22 August 2026, 2:02 PM
State	Finished
Completed on	Saturday, 22 August 2026, 2:04 PM
Time taken	1 min 7 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int count = 0;

    scanf("%d", &n);

    for(int i = 1; i <= n; ++i)
    {
        count++;
        count++;

        if(n % i == 0)
        {
            count++;
        }
    }

    count++;

    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔

	Input	Expected	Got	
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Santhosh S (dob 14.02.2008) 251801299-2025-AIDS

S2

Started on	Saturday, 22 August 2026, 2:04 PM
State	Finished
Completed on	Saturday, 22 August 2026, 2:09 PM
Time taken	4 mins 56 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int c = 0;

    scanf("%d", &n);

    int i = n / 2;
    c++;

    while(1)
    {
        c++;
        if(i >= n)
            break;

        int j = 1;

        while(1)
        {
            c++;
            if(j >= n)
                break;

            int k = 1;

            while(1)
            {
                c++;
                if(k >= n)
                    break;

                c++;
                k = 2 * k;
            }
```

```
            j = 2 * j;
        }

        i++;
    }

    printf("%d", c);

    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Santhosh S (dob 14.02.2008) 251801299-2025-AIDS

S2

Started on	Saturday, 22 August 2026, 2:09 PM
State	Finished
Completed on	Saturday, 22 August 2026, 2:26 PM
Time taken	16 mins 16 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main()
{
    int n;
    int rev = 0, remainder;
    int count=0;

    scanf("%d", &n);
    count++;
    count++;


    while(n != 0)
    {
        count++;
        remainder = n % 10;

        count++;
        rev = rev * 10 + remainder;

        count++;
        n = n / 10;

        count++;
    }
       count++;




    printf("%d", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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